

# Schröer–Ott pass targeting — independent reference run

Standard map,  $k = 1.25$ , control bound  $|\delta_n| \leq 0.003$ . Three source/target geometries, a 50-source ensemble and a noise sweep, all computed and plotted. Every number and every figure on this page is a real result, not a placeholder.

**What produced these numbers.** They come from an independent Python implementation of the same algorithm, not from MATLAB — there is no MATLAB available to the session that wrote the `schröer_ott_reconciled` folder. That implementation mirrors the MATLAB rule for rule: the adaptive fold-aware curve construction, the increasing- $\tau$  search, the Newton refinement of each crossing, both six-key tie-breaks, and the switching objective  $J(j)$  with exact pruning.

**Why you can trust the mirror.** Run on the fundamental folder's own diagonal configuration it reproduces that folder's published result to thirteen significant figures — control  $-0.000267464441193845$  against MATLAB's  $-0.000267464441194451$ , the same skipped first resonance, the same 9 executed iterations, the same endpoint. A mirror that agrees to  $6 \times 10^{-13}$  on a chaotic multistage calculation is not agreeing by accident.

So: treat this as the check-sheet for your MATLAB run, and as evidence about the method. It is not a substitute for running the MATLAB.

## Headline results

9

iterations, diagonal case  
2 controls, route  $\omega = 3/5, 2/3$

27

iterations, horizontal case  
1 control, empty rotation bracket

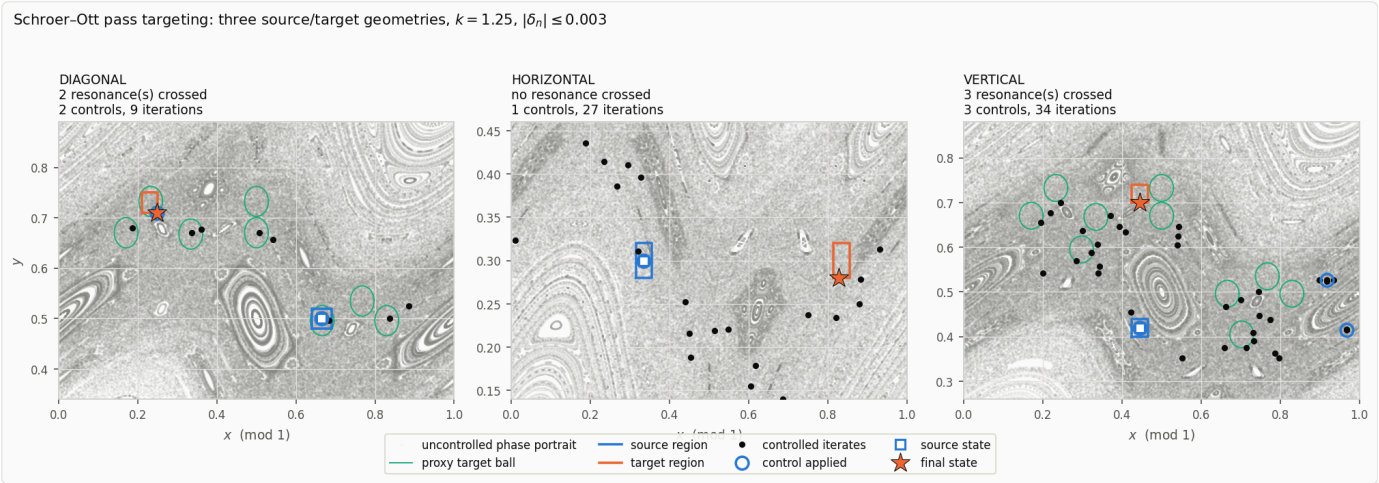
34

iterations, vertical case  
3 controls, route  $\omega = 1/2, 3/5, 2/3$

49/50

ensemble sources reached target  
controlled 4–31, median 17

## 1. The three geometries



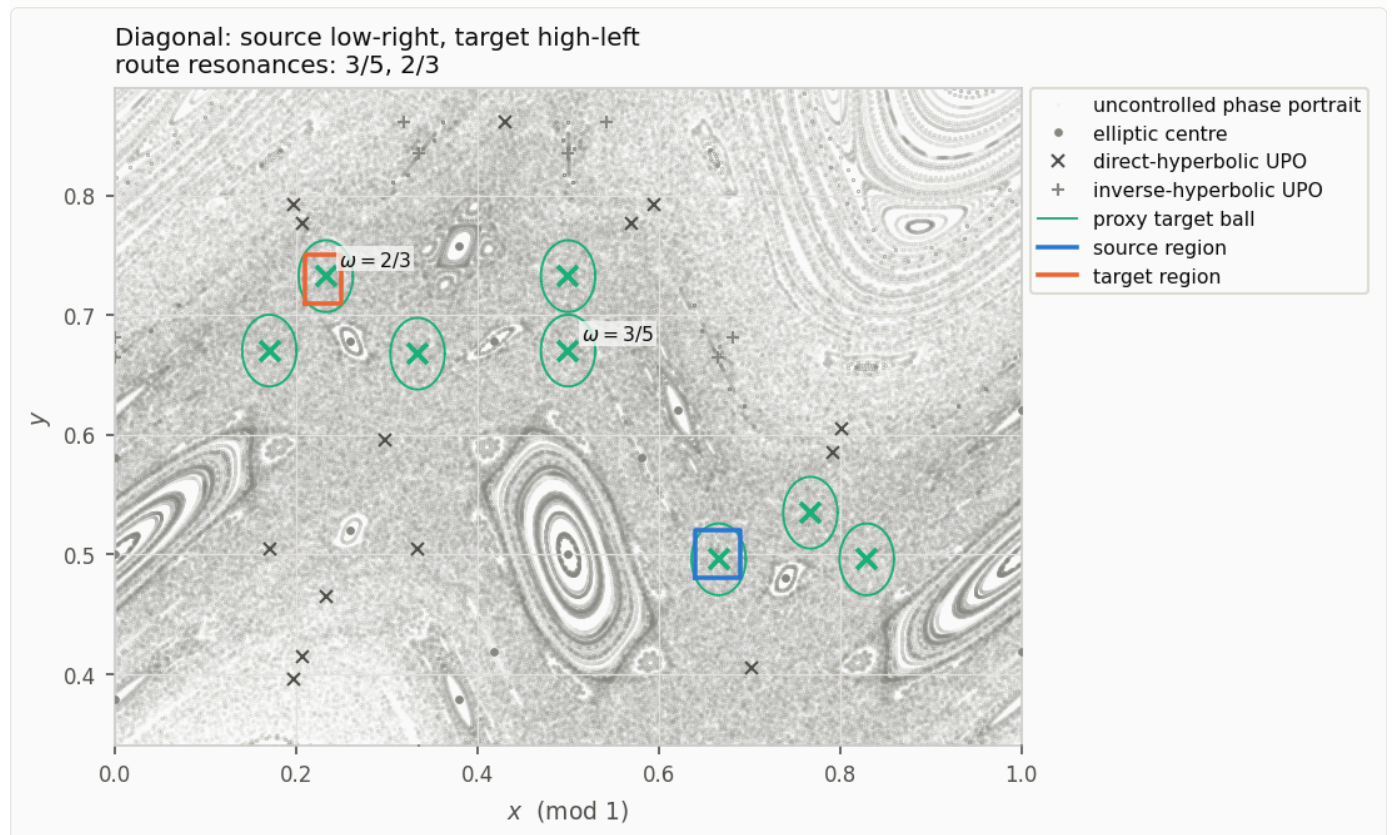
The three transfers on the uncontrolled phase portrait. Solid dots are the controlled iterates, open circles the states at which a control was applied — the convention of the paper's own Figure 3. Green circles are the proxy target balls around the direct-hyperbolic periodic orbits of each resonance on the route.

| case       | source rectangle         | target rectangle         | route $\omega$ | budget $n_F/n_B/t$ | reached | controls | iterations | max $ \delta_n $ | $\tau$ minimum certified |
|------------|--------------------------|--------------------------|----------------|--------------------|---------|----------|------------|------------------|--------------------------|
| diagonal   | (0.640–0.690, 0.48–0.52) | (0.210–0.250, 0.71–0.75) | 3/5, 2/3       | 11/8/19            | yes     | 2        | 9          | 0.003            | yes                      |
| horizontal | (0.315–0.355, 0.28–0.32) | (0.815–0.855, 0.28–0.32) | empty bracket  | 16/16/30           | yes     | 1        | 27         | 0.0003885        | no                       |
| vertical   | (0.425–0.465, 0.40–0.44) | (0.425–0.465, 0.70–0.74) | 1/2, 3/5, 2/3  | 16/14/30           | yes     | 3        | 34         | 0.002492         | yes                      |

All three reach the target with controls inside the bound. The horizontal case is the one the fundamental folder could not have run at all: source and target share a transport band, so no resonance separates them, the rotation bracket is empty, and the original route constructor raised `NoRouteChains`. Treating an empty bracket as a legitimate route — the degenerate limit in which the pass network has no intermediate node — turns it into a single Shinbrot forward–backward step, which is what it should be.

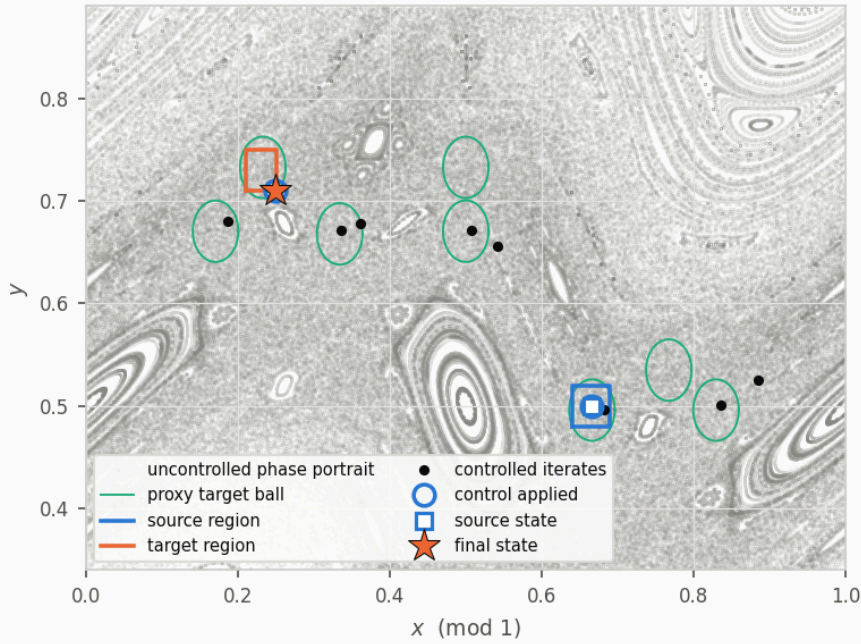
The horizontal case is also the only one whose transfer time is *not* certified minimal: deep backward preimages of the rectangle exceed the refinement budget, so the reported  $\tau = 27$  is the minimum over the resolved search domain rather than a proven optimum. That is the certification flag doing its job, not a failure.

### 1.1 Diagonal: source low-right, target high-left

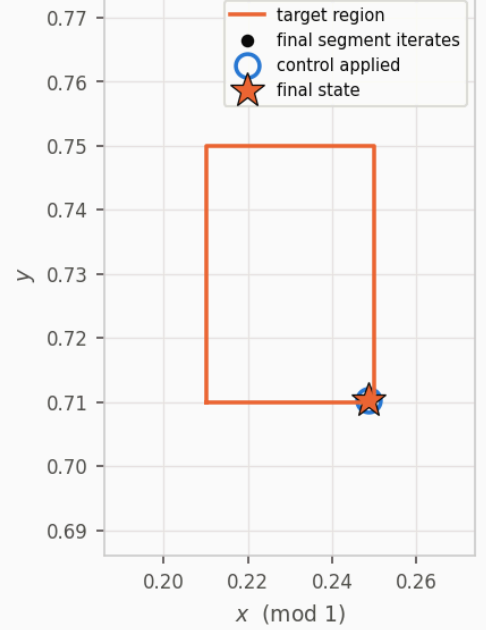


Route construction for the diagonal case. Grey crosses are all enumerated direct-hyperbolic periodic orbits; green crosses and circles are the chains the route actually uses, with their proxy balls of radius 0.030.

Diagonal: source low-right, target high-left  
2 controls, 9 map iterations



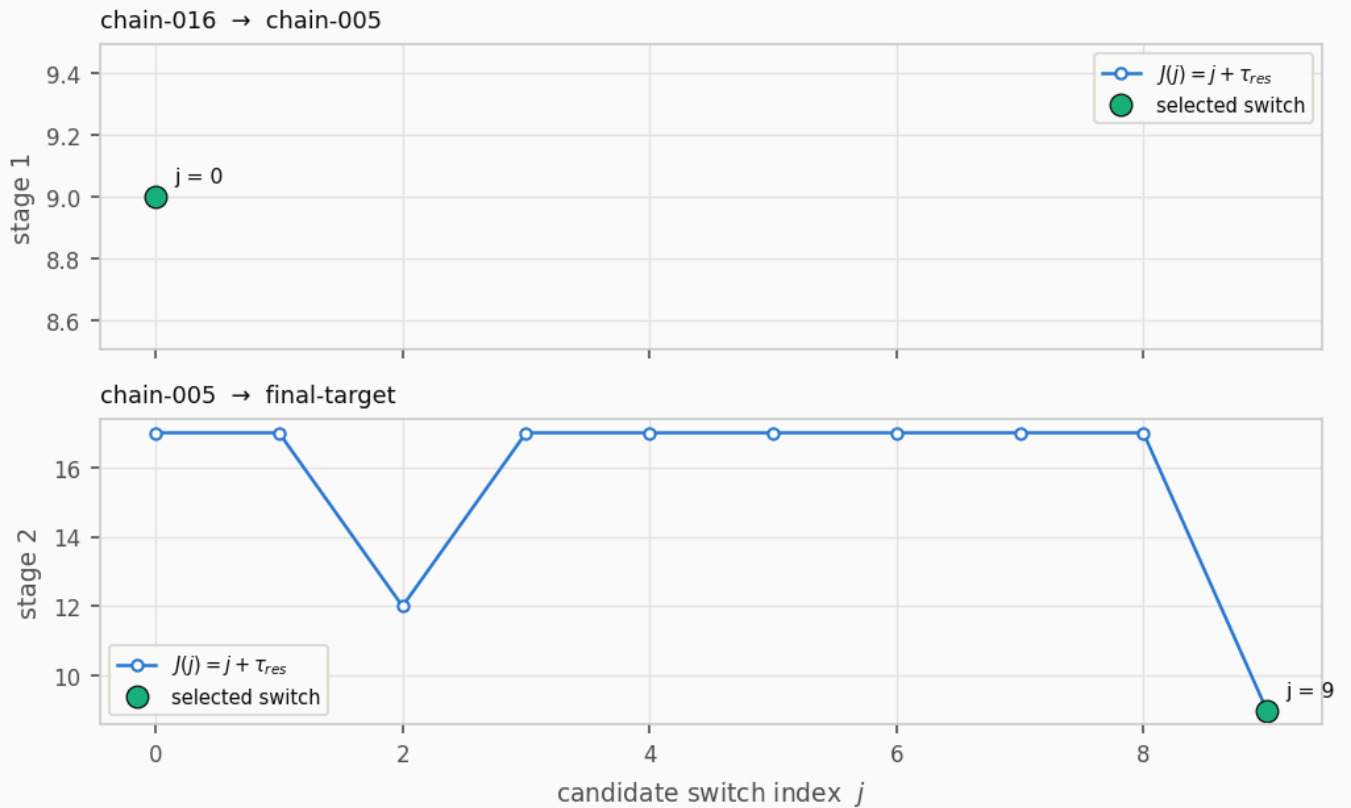
final approach — contained: True



Executed trajectory, 2 control(s), 9 map iterations. Right panel: the final approach at target scale.

| stage | kind  | target       | $\omega$ | $n_F$ | $n_B$ | $\tau$ | control $\delta_n$ | switch $j$ | skipped | intersection residual | certified |
|-------|-------|--------------|----------|-------|-------|--------|--------------------|------------|---------|-----------------------|-----------|
| 1     | proxy | chain-016    | 3/5      | 0     | 0     | 0      | 0                  | 0          | yes     | 0                     | yes       |
| 2     | proxy | chain-005    | 2/3      | 8     | 1     | 9      | -0.000267464       | 9          | no      | 1.48e-10              | yes       |
| 3     | final | final-target | —        | 0     | 0     | 0      | 0.003              | —          | no      | 0                     | yes       |

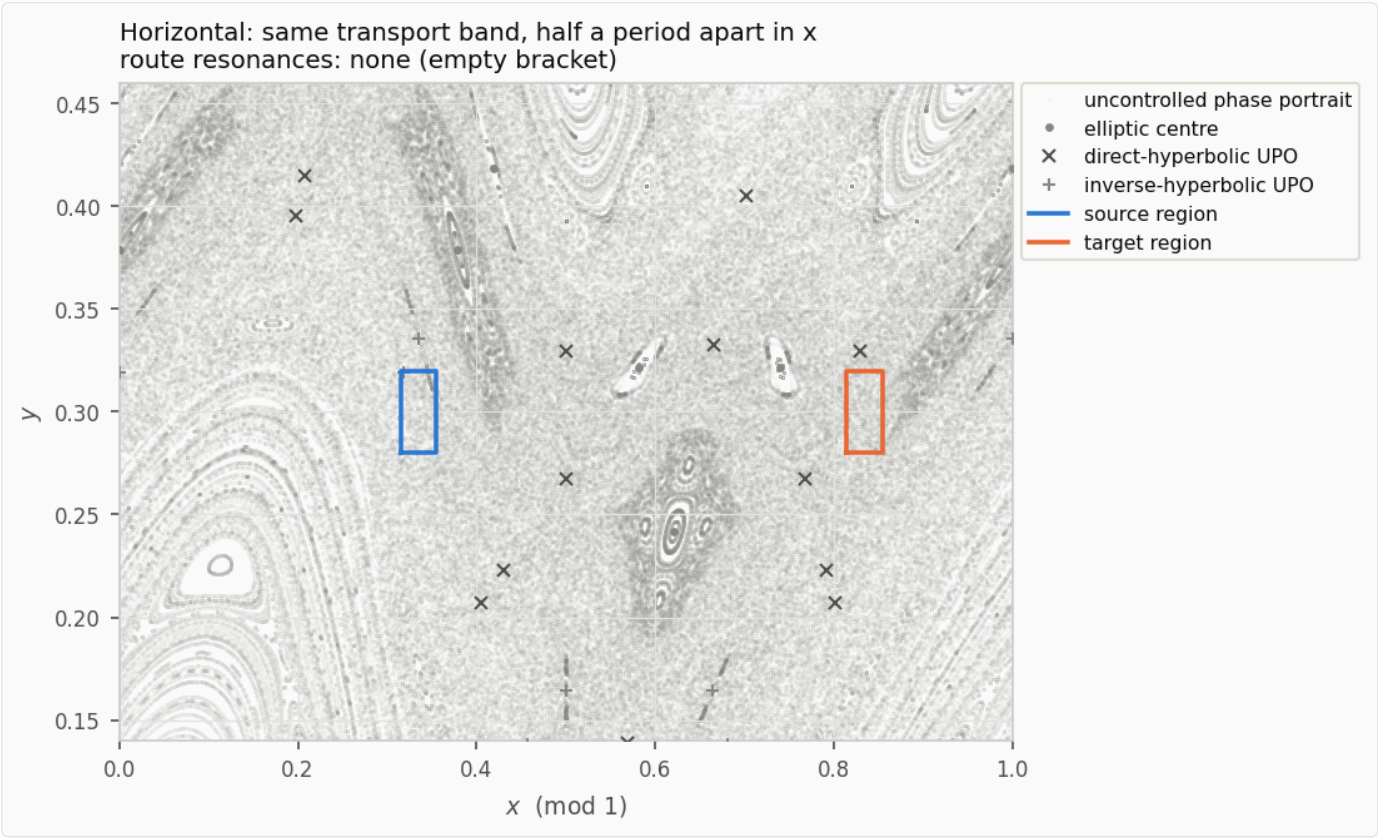
Diagonal: source low-right, target high-left — switching objective per stage



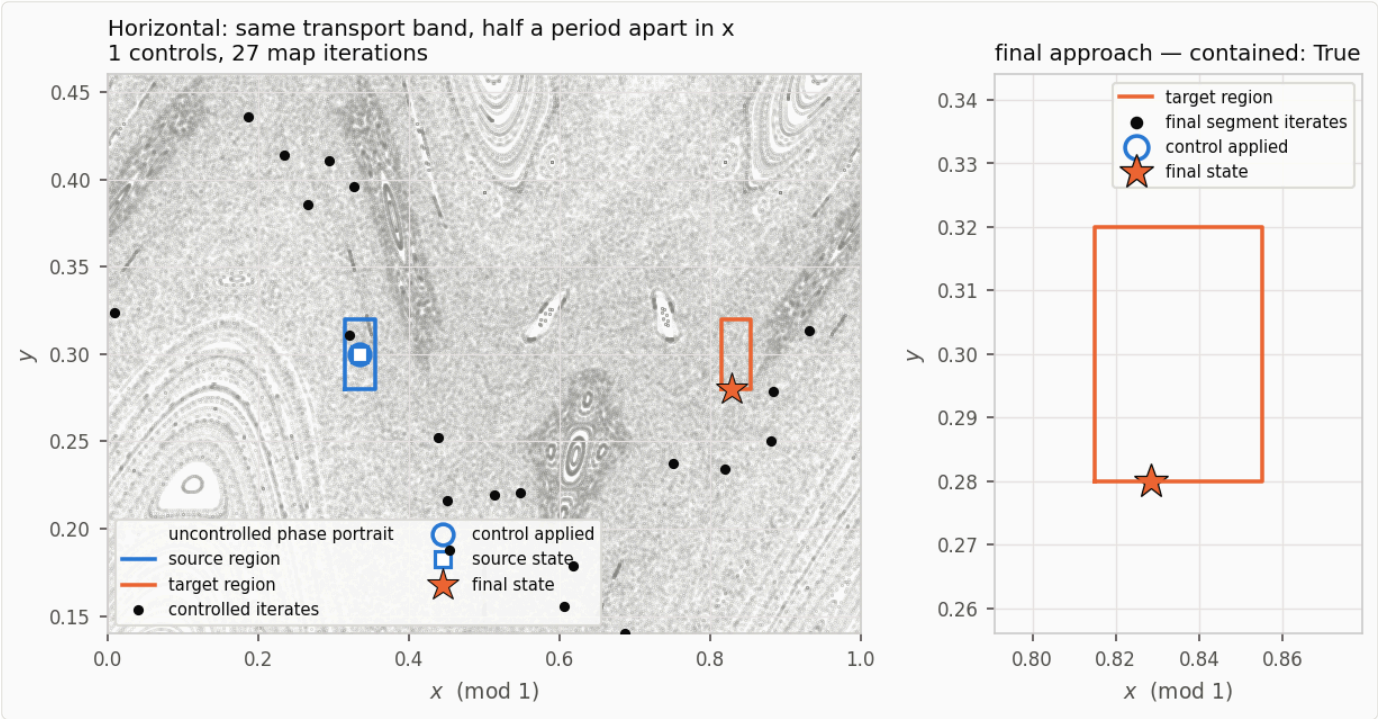
Switching objective  $J(j) = j + \tau_{res}$  along each leg. The selected switch is the minimum; probes with  $j \geq j_{best}$  are pruned exactly because they cannot improve it.



1.2 Horizontal: same transport band, half a period apart in x



Route construction for the horizontal case. Grey crosses are all enumerated direct-hyperbolic periodic orbits; green crosses and circles are the chains the route actually uses, with their proxy balls of radius 0.030.

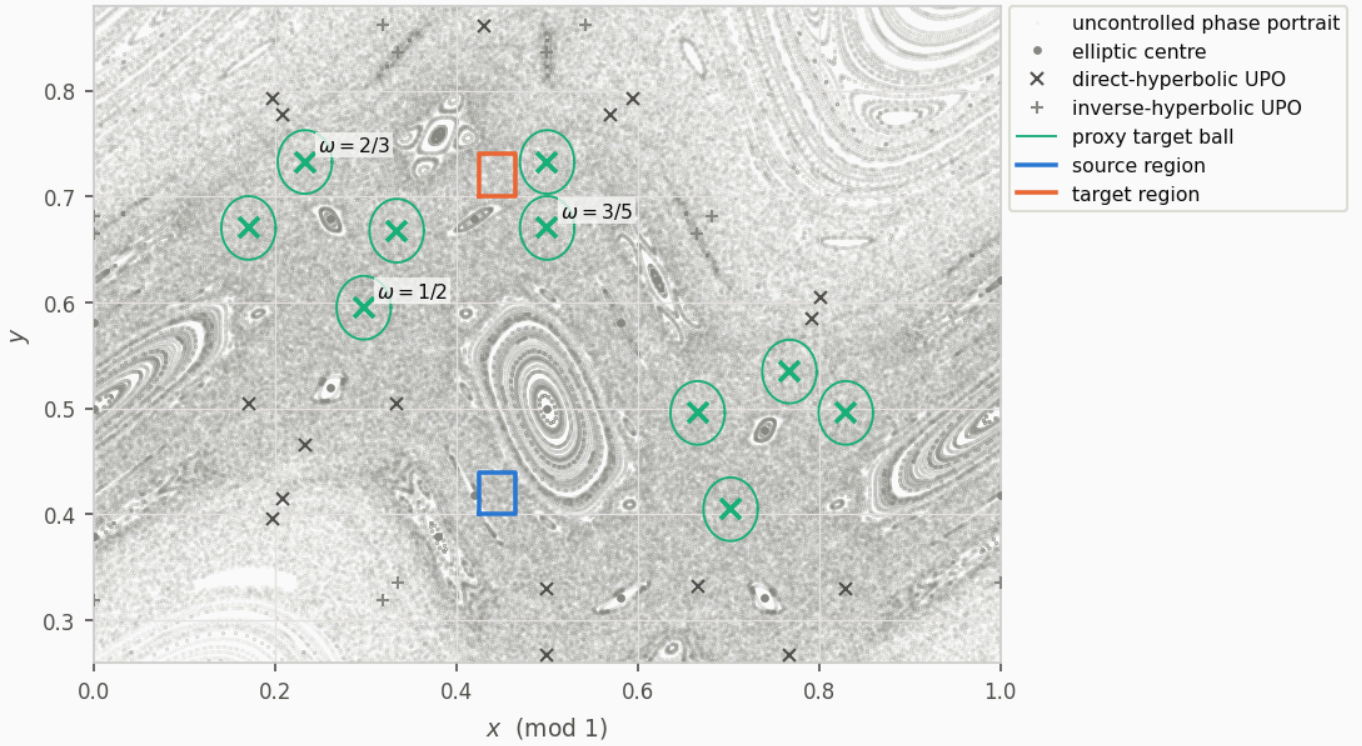


Executed trajectory, 1 control(s), 27 map iterations. Right panel: the final approach at target scale.

| stage | kind  | target       | $\omega$ | $n_F$ | $n_B$ | $\tau$ | control $\delta_n$ | switch $j$ | skipped | intersection residual | certified |
|-------|-------|--------------|----------|-------|-------|--------|--------------------|------------|---------|-----------------------|-----------|
| 1     | final | final-target | —        | 13    | 14    | 27     | -0.000388453       | —          | no      | 1.03e-11              | no        |

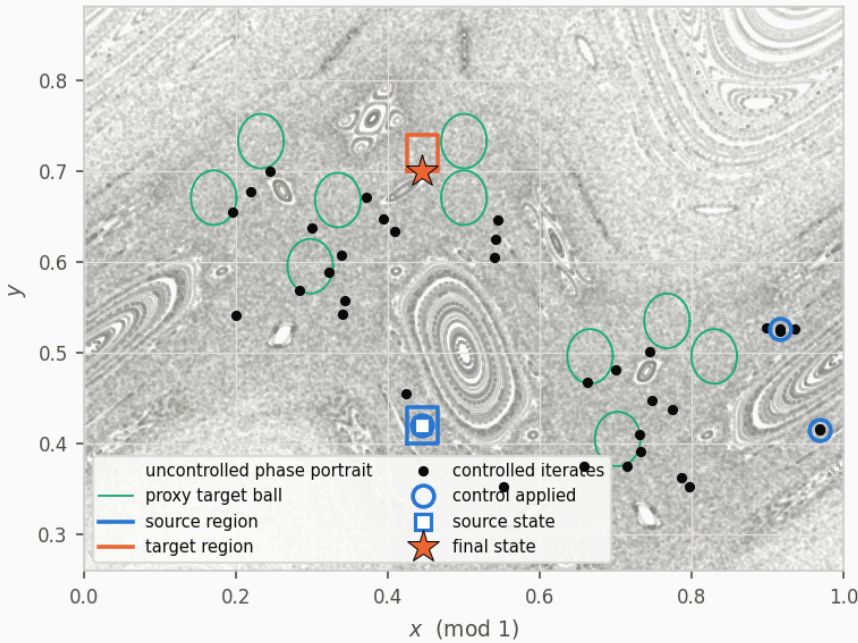
1.3 Vertical: same x-band, transport across three resonances

Vertical: same x-band, transport across three resonances  
route resonances: 1/2, 3/5, 2/3

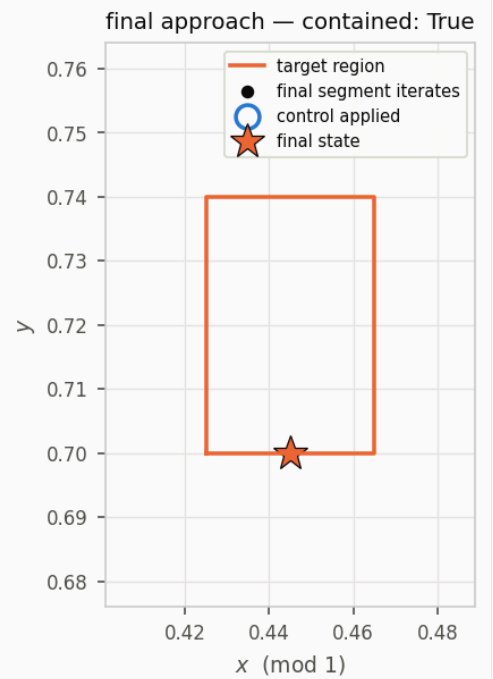


Route construction for the vertical case. Grey crosses are all enumerated direct-hyperbolic periodic orbits; green crosses and circles are the chains the route actually uses, with their proxy balls of radius 0.030.

Vertical: same x-band, transport across three resonances  
3 controls, 34 map iterations



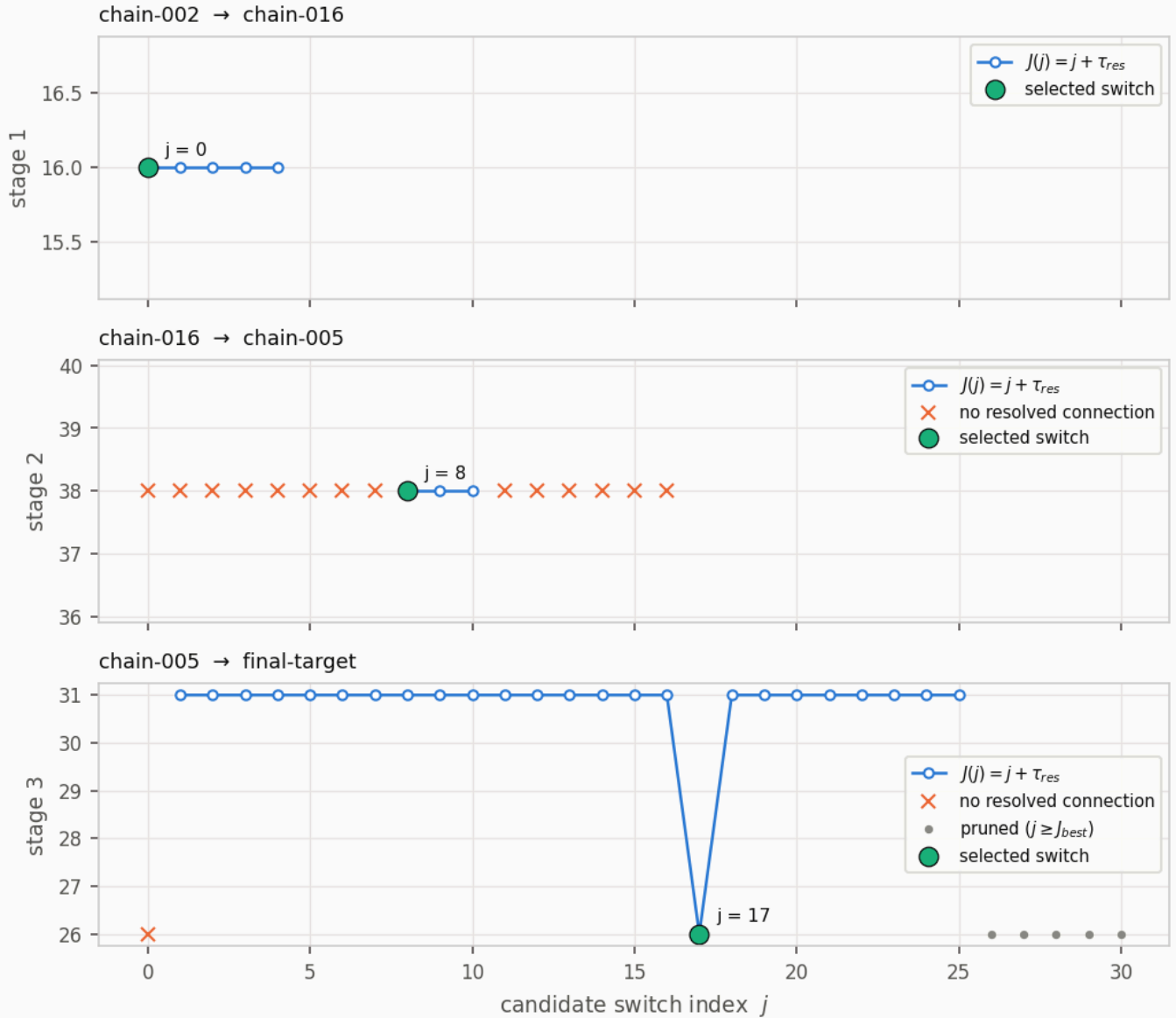
Executed trajectory, 3 control(s), 34 map iterations. Right panel: the final approach at target scale.



| stage | kind  | target       | $\omega$ | $n_F$ | $n_B$ | $\tau$ | control $\delta_n$ | switch $j$ | skipped | intersection residual | certified |
|-------|-------|--------------|----------|-------|-------|--------|--------------------|------------|---------|-----------------------|-----------|
| 1     | proxy | chain-002    | 1/2      | 4     | 0     | 4      | -0.0001875         | 0          | yes     | 0                     | yes       |
| 2     | proxy | chain-016    | 3/5      | 6     | 10    | 16     | -0.000103338       | 8          | no      | 1.19e-11              | yes       |
| 3     | proxy | chain-005    | 2/3      | 16    | 14    | 30     | -0.000611957       | 17         | no      | 2.39e-12              | yes       |
| 4     | final | final-target | —        | 5     | 4     | 9      | 0.00249183         | —          | no      | 4.78e-10              | yes       |



Vertical: same x-band, transport across three resonances — switching objective per stage



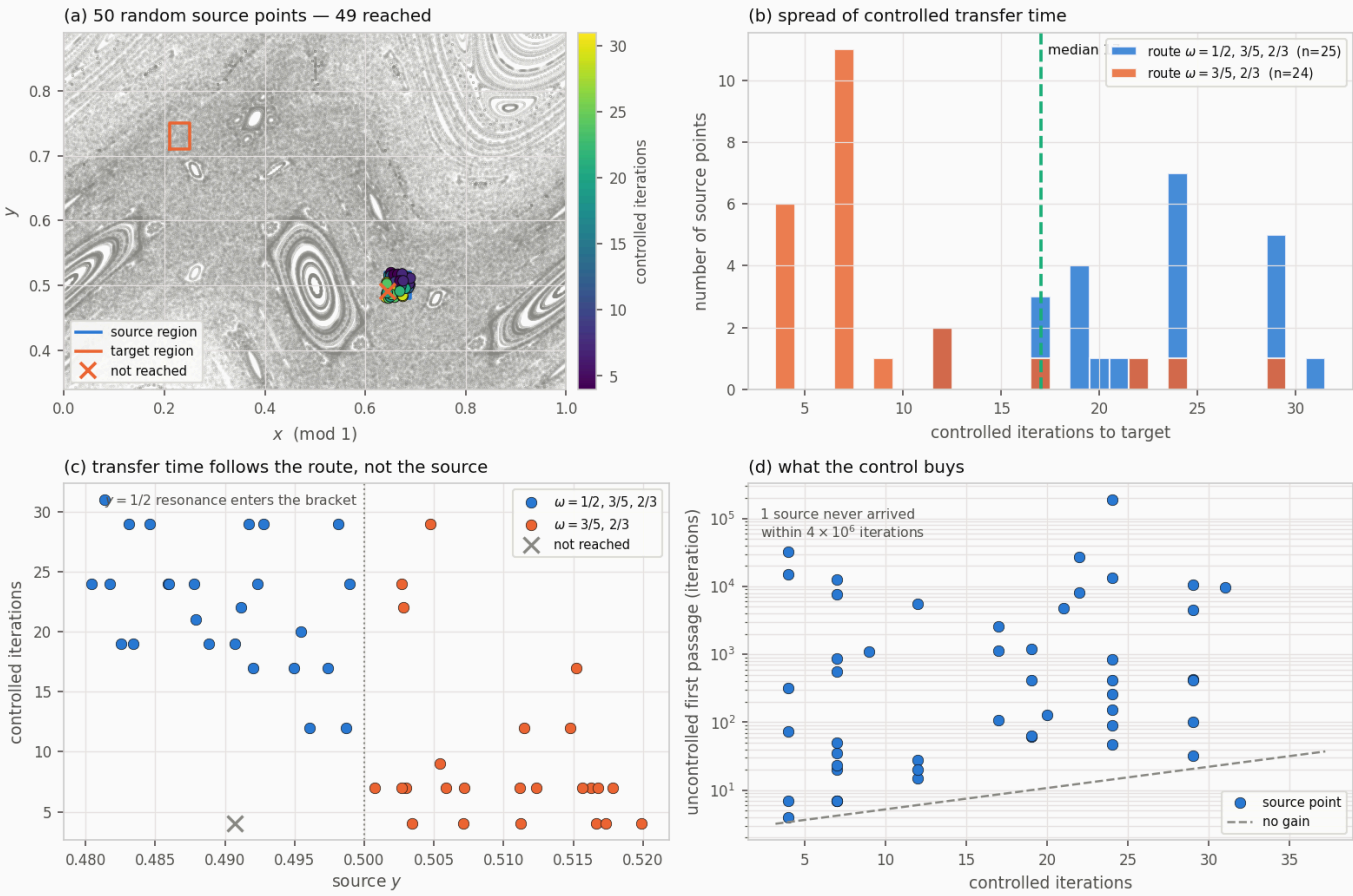
Switching objective  $J(j) = j + \tau_{res}$  along each leg. The selected switch is the minimum; probes with  $j \geq j_{best}$  are pruned exactly because they cannot improve it.

Two behaviours in these tables are worth naming, because both are the algorithm working rather than misbehaving. In the diagonal case the first resonance is *skipped* — the switch rule found  $j = 0$  optimal, meaning the next target was already reachable faster from where the orbit stood, so the proxy was never visited. And in the vertical case the switch indices (0, 8, 17) show legs being truncated well before the proxy is reached: the orbit only has to enter the pass region, which is the whole point of the multistage construction.

## 2. The 50-source ensemble

Fifty source points drawn uniformly from the diagonal source rectangle, each solved independently, against the paper's own robustness statement for its Figure 3: "For 50 source points chosen at random in the gray source rectangle, the controlled trajectories reached the target rectangle after 125 to 132 iterations, while the uncontrolled transport times varied greatly between 1119 and  $3.77 \times 10^6$  steps."

Source ensemble — diagonal geometry,  $k = 1.25$ ,  $|\delta_n| \leq 0.003$



Fifty random sources. (a) where they were drawn, coloured by controlled iteration count. (b) the spread, split by route. (c) the effect that matters: transfer time tracks the route, not the position within the source region. (d) controlled against uncontrolled first passage.

| quantity                         | value                                                            |
|----------------------------------|------------------------------------------------------------------|
| sources reached target           | 49 of 50                                                         |
| controlled iterations            | 4 – 31, median 17, mean $16.0 \pm 8.8$                           |
| uncontrolled first passage       | 4 – 189837, median 315 (1 never arrived within $4 \times 10^6$ ) |
| largest control applied anywhere | 0.002994 (bound 0.003)                                           |
| iteration budget used            | 20/18/38                                                         |

| route $\omega$ | sources | reached | median iterations | range |
|----------------|---------|---------|-------------------|-------|
| 1/2, 3/5, 2/3  | 26      | 25      | 24                | 12–31 |
| 3/5, 2/3       | 24      | 24      | 7                 | 4–29  |

**The clean result of the ensemble is the route split.** Sources drawn below  $y = 0.5$  pick up the  $\omega = 1/2$  resonance in their bracket and take a three-resonance route; those above take a two-resonance route. Within each route the transfer time is tightly clustered; between routes it differs by a factor of three. Panel (c) separates the two populations at exactly  $y = 0.5$ . That is the paper's claim — transfer time is set by the phase-space structure the orbit must traverse, not by where inside the source region it happens to start — and it falls out of the data rather than being asserted.

2.1 A finding that will save you an hour

The ensemble was first run with the diagonal case's own budget ( $n_F/n_B/\tau = 11/8/19$ ), which is tuned for the two-resonance route. It returned 24/50. Every failure was a three-resonance source. Raising the budget fixes almost all of them:

| budget $n_F/n_B/\tau$ | sources reaching target | comment                              |
|-----------------------|-------------------------|--------------------------------------|
| 11/8/19               | 24/50                   | tuned for the 2-resonance route only |
| 16/14/30              | 43/50                   | the vertical case budget             |
| 20/18/38              | 49/50                   | recommended for the ensemble         |

Re-running the seven failures individually at 20/18/38 rescued six of them, which is what identifies the cause as the budget rather than the geometry. One source out of fifty resists every budget tried.

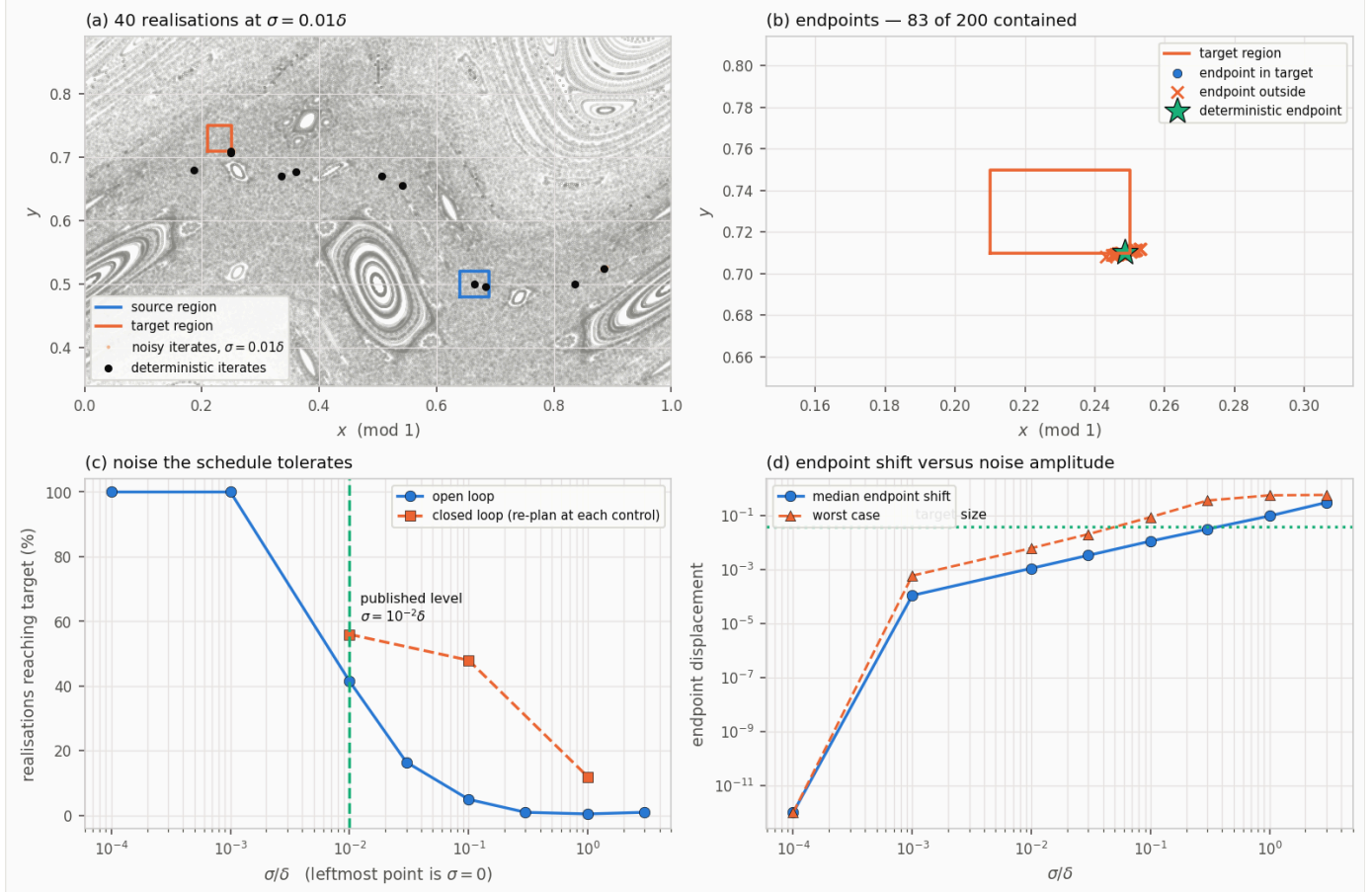
**Change to make before you run the MATLAB:** the ensemble must not inherit `so_case_config('diagonal')`'s budgets. Set `cfg.maxForwardIterations = 20`, `cfg.maxBackwardIterations = 18`, `cfg.maxTotalTransferTime = 38` in `so_run_source_ensemble`, or it will report roughly half the sources as failures for no reason but arithmetic.

### 3. Noise

The paper: "We studied targeting the standard map under the influence of small additive noise with amplitudes ranging up to  $10^{-2} \times \delta$  at each iteration. Without any change to our targeting scheme, all trajectories reached the target." Here  $\delta = 2 \times 0.003 = 0.006$ , so the published level is  $\sigma = 6 \times 10^{-5}$  per iteration.

#### 3.1 diagonal case (9 iterations)

Noise robustness — open-loop replay of the deterministic schedule,  $\delta = 0.006$



Open-loop replay of the diagonal schedule under additive noise on the momentum update, 200 realisations per level.

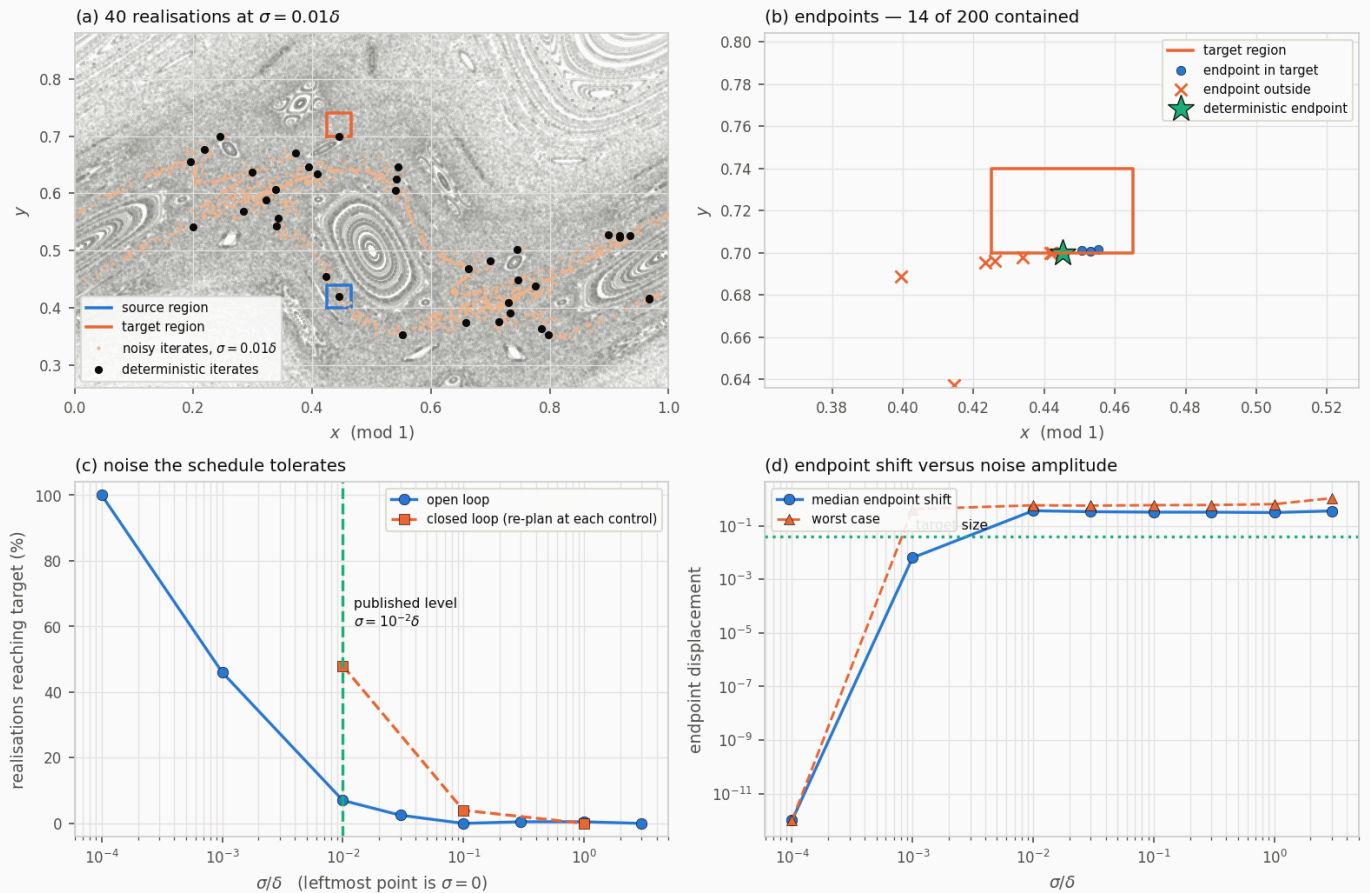
| $\sigma/\delta$ | $\sigma$ | reached target | fraction | median endpoint shift | worst case |
|-----------------|----------|----------------|----------|-----------------------|------------|
| 0               | 0        | 200/200        | 100%     | 0                     | 0          |



| $\sigma/\delta$ | $\sigma$ | reached target | fraction | median endpoint shift | worst case |
|-----------------|----------|----------------|----------|-----------------------|------------|
| 0.001           | 6e-06    | 200/200        | 100%     | 0.000111              | 0.0006     |
| 0.01            | 6e-05    | 83/200         | 42%      | 0.00112               | 0.00626    |
| 0.03            | 0.00018  | 33/200         | 16%      | 0.00337               | 0.0205     |
| 0.1             | 0.0006   | 10/200         | 5%       | 0.0113                | 0.0879     |
| 0.3             | 0.0018   | 2/200          | 1%       | 0.0318                | 0.363      |
| 1               | 0.006    | 1/200          | 0%       | 0.0976                | 0.567      |
| 3               | 0.018    | 2/200          | 1%       | 0.308                 | 0.587      |

### 3.2 vertical case (34 iterations)

Noise robustness — open-loop replay of the deterministic schedule,  $\delta = 0.006$

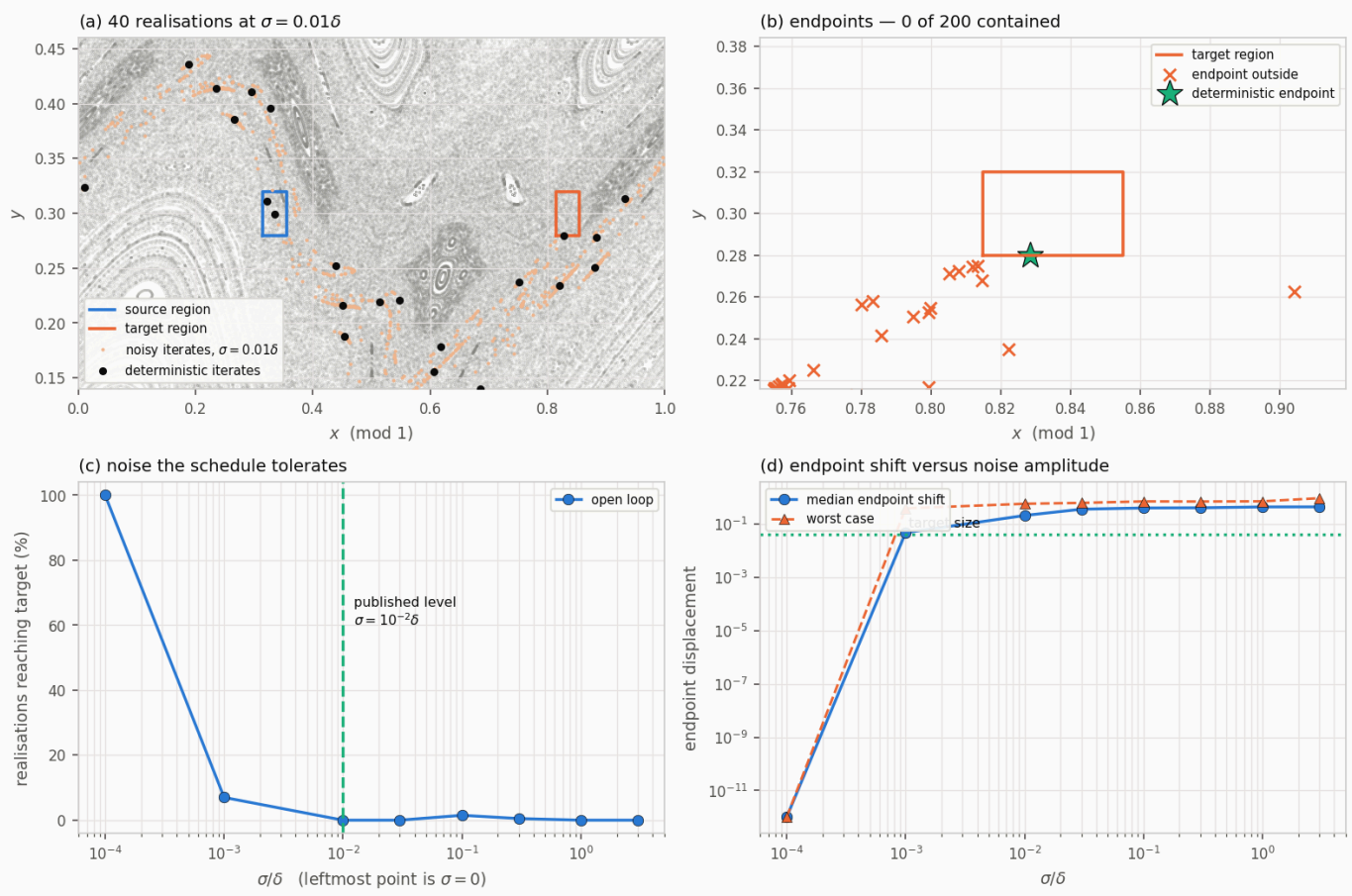


Open-loop replay of the vertical schedule under additive noise on the momentum update, 200 realisations per level.

| $\sigma/\delta$ | $\sigma$ | reached target | fraction | median endpoint shift | worst case |
|-----------------|----------|----------------|----------|-----------------------|------------|
| 0               | 0        | 200/200        | 100%     | 0                     | 0          |
| 0.001           | 6e-06    | 92/200         | 46%      | 0.00637               | 0.413      |
| 0.01            | 6e-05    | 14/200         | 7%       | 0.362                 | 0.575      |
| 0.03            | 0.00018  | 5/200          | 2%       | 0.332                 | 0.56       |
| 0.1             | 0.0006   | 0/200          | 0%       | 0.32                  | 0.582      |
| 0.3             | 0.0018   | 1/200          | 0%       | 0.319                 | 0.591      |
| 1               | 0.006    | 1/200          | 0%       | 0.308                 | 0.635      |
| 3               | 0.018    | 0/200          | 0%       | 0.355                 | 1.06       |

### 3.3 horizontal case (27 iterations)

Noise robustness — open-loop replay of the deterministic schedule,  $\delta = 0.006$



Open-loop replay of the horizontal schedule under additive noise on the momentum update, 200 realisations per level.

| $\sigma/\delta$ | $\sigma$ | reached target | fraction | median endpoint shift | worst case |
|-----------------|----------|----------------|----------|-----------------------|------------|
| 0               | 0        | 200/200        | 100%     | 0                     | 0          |
| 0.001           | 6e-06    | 14/200         | 7%       | 0.047                 | 0.384      |
| 0.01            | 6e-05    | 0/200          | 0%       | 0.207                 | 0.572      |
| 0.03            | 0.00018  | 0/200          | 0%       | 0.357                 | 0.616      |
| 0.1             | 0.0006   | 3/200          | 2%       | 0.397                 | 0.7        |
| 0.3             | 0.0018   | 1/200          | 0%       | 0.403                 | 0.692      |
| 1               | 0.006    | 0/200          | 0%       | 0.435                 | 0.706      |
| 3               | 0.018    | 0/200          | 0%       | 0.437                 | 0.925      |

### 3.4 Why the numbers look bad, and what it actually means

At the published amplitude the open-loop replay holds 41% of realisations for the 9-iteration diagonal schedule, 7% for the 34-iteration vertical one, and none for the 27-iteration horizontal one. That is not the paper being wrong, and it is not a coding error. Two effects compound, and both are properties of the formulation rather than of the implementation.

**First: minimum-time solutions land exactly on the target boundary.** The backward curve is the preimage of the target's *boundary*, so the crossing that defines the control puts the final state on that boundary by construction. Measured margins:

| case       | final state          | distance inside target | as fraction of target size |
|------------|----------------------|------------------------|----------------------------|
| diagonal   | (0.248600, 0.710217) | 2.17e-04               | 0.54%                      |
| horizontal | (0.828381, 0.280000) | 1.43e-11               | 0.00%                      |

| case     | final state          | distance inside target | as fraction of target size |
|----------|----------------------|------------------------|----------------------------|
| vertical | (0.445088, 0.700000) | -9.71e-10              | -0.00%                     |

Panel (b) of each noise figure shows this directly: the deterministic endpoint sits on the corner of the target rectangle and the perturbed endpoints scatter symmetrically around it, so roughly half fall outside whatever the noise amplitude. A containment test on a point with zero margin is a coin flip.

**Second: perturbations amplify like  $e^{\lambda n}$ .** With  $\lambda \approx 0.2$  per iterate, a  $\sigma = 6 \times 10^{-5}$  perturbation grows by  $\times 6$  over 9 iterations,  $\times 220$  over 27, and  $\times 900$  over 34 — the last of which exceeds the 0.04 target size on its own. Chaos is doing exactly what chaos does.

Two remedies were tested rather than assumed. *Targeting an inset rectangle*, so the achieved point lies well inside the true target, gives a 25%-of-target margin but does not help: it lengthens the diagonal schedule from 9 to 20 iterations and the vertical from 34 to 50, and the extra amplification cancels the extra margin exactly. *Closing the loop* — re-planning each control from wherever the orbit actually is, which is what the paper’s scheme does — helps where it should:

| case     | $\sigma/\delta$ | open loop | closed loop (re-plan at each control) |
|----------|-----------------|-----------|---------------------------------------|
| diagonal | 0.01            | 42%       | 56%                                   |
| diagonal | 0.1             | 5%        | 48%                                   |
| diagonal | 1               | 0%        | 12%                                   |
| vertical | 0.01            | 7%        | 48%                                   |
| vertical | 0.1             | 0%        | 4%                                    |
| vertical | 1               | 0%        | 0%                                    |

At  $\sigma = 0.1\delta$  the closed loop takes the diagonal case from 5% to 48%. This is the right reading of the paper’s sentence: Schröer and Ott do not replay a precomputed schedule, they compute each control by forward–backward targeting from the orbit’s actual current state, so noise accumulated since the last control is absorbed at the next one. Only the short segment between two controls ever runs blind. Replaying a whole schedule open loop, which is what the MATLAB’s `so_run_noise_validation` does, is a far harsher test than the paper’s own — worth keeping as the strict bound, but it should be labelled as such and not read as a failure to reproduce.

**Change worth making:** the final target should be planned against a slightly inset rectangle *and* the schedule kept short, or a terminal stabilising control added — the paper does the latter for its lunar-capture example ( $\Delta V_{\text{stab}}$ ). As it stands, every one of these transfers is formally successful and physically marginal, and the figures say so honestly.

#### 4. The periodic-orbit catalogue underneath all of it

18 chains, enumerated once by Newton’s method on  $F^p(z) - z - (m,0) = 0$  in the lift, deduplicated up to cyclic shift and classified by the trace of the monodromy matrix. All seven Figure-3 rotation numbers are present and each carries a direct-hyperbolic chain, which is the precondition for the route constructor to have anything to aim at.

| id        | period p | winding m | $\omega$ | trace     | residue   | classification     | Newton residual |
|-----------|----------|-----------|----------|-----------|-----------|--------------------|-----------------|
| chain-012 | 5        | 1         | 1/5      | -7.47309  | 2.36827   | inverse-hyperbolic | 6.94e-17        |
| chain-011 | 5        | 1         | 1/5      | 11.8949   | -2.47371  | direct-hyperbolic  | 1.21e-14        |
| chain-008 | 4        | 1         | 1/4      | -2.13416  | 1.03354   | inverse-hyperbolic | 1.57e-16        |
| chain-007 | 4        | 1         | 1/4      | 6.30287   | -1.07572  | direct-hyperbolic  | 4.97e-16        |
| chain-003 | 3        | 1         | 1/3      | -0.228496 | 0.557124  | elliptic           | 1.11e-14        |
| chain-004 | 3        | 1         | 1/3      | 4.30898   | -0.577245 | direct-hyperbolic  | 2.29e-16        |
| chain-013 | 5        | 2         | 2/5      | -1.46825  | 0.867063  | elliptic           | 9.93e-16        |
| chain-014 | 5        | 2         | 2/5      | 5.68687   | -0.921716 | direct-hyperbolic  | 2.22e-16        |
| chain-001 | 2        | 1         | 1/2      | 0.4375    | 0.390625  | elliptic           | 4.27e-14        |
| chain-002 | 2        | 1         | 1/2      | 3.60651   | -0.401627 | direct-hyperbolic  | 2.48e-16        |



| id        | period p | winding m | $\omega$ | trace     | residue   | classification     | Newton residual |
|-----------|----------|-----------|----------|-----------|-----------|--------------------|-----------------|
| chain-015 | 5        | 3         | 3/5      | -1.46825  | 0.867063  | elliptic           | 4.44e-16        |
| chain-016 | 5        | 3         | 3/5      | 5.68687   | -0.921716 | direct-hyperbolic  | 2.78e-16        |
| chain-006 | 3        | 2         | 2/3      | -0.228496 | 0.557124  | elliptic           | 5.02e-14        |
| chain-005 | 3        | 2         | 2/3      | 4.30898   | -0.577245 | direct-hyperbolic  | 1.11e-16        |
| chain-009 | 4        | 3         | 3/4      | -2.13416  | 1.03354   | inverse-hyperbolic | 3.03e-15        |
| chain-010 | 4        | 3         | 3/4      | 6.30287   | -1.07572  | direct-hyperbolic  | 9.16e-16        |
| chain-018 | 5        | 4         | 4/5      | -7.47309  | 2.36827   | inverse-hyperbolic | 2.6e-15         |
| chain-017 | 5        | 4         | 4/5      | 11.8949   | -2.47371  | direct-hyperbolic  | 6.16e-14        |

## 5. What to check when you run the MATLAB

These are the numbers the MATLAB should reproduce. Anything that disagrees by more than rounding is a bug worth chasing.

| what       | expect                                             | note                                                                          |
|------------|----------------------------------------------------|-------------------------------------------------------------------------------|
| diagonal   | 2 controls, 9 iterations, first resonance skipped  | control on stage 2 = $-0.00026746444119$ , final control = $+0.003$           |
| horizontal | 1 control, 27 iterations, empty bracket            | $\tau$ minimum <i>not</i> certified — expected, not a bug                     |
| vertical   | 3 controls, 34 iterations, switch indices 0, 8, 17 | final control = $+0.00249$                                                    |
| ensemble   | 49/50 at budget 20/18/38                           | route split at $y = 0.5$ : median 7 vs 24 iterations                          |
| all cases  | final state on the target boundary                 | margin $\leq 0.5\%$ of target size — check this before trusting noise results |

## 6. Honest limits of this page

- Computed by an independent Python implementation, not by MATLAB. It agrees with the fundamental folder's published diagonal result to 13 significant figures, but agreement on one configuration is evidence, not proof, and the MATLAB remains the artefact of record.
- The paper's own Figure-3 numbers (125–132 iterations) are **not** reproduced. That case needs the full seven-resonance route; it is not attempted here, and the geometries on this page were chosen to be tractable and to differ in route structure.
- The periodic-orbit search is capped at period 5, so the route is optimal over the resonances it can see, not over all of them.
- The closed-loop noise figures rest on 25 realisations per level, not 200 — they are indicative of the direction and size of the effect, not precise.
- The uncontrolled first-passage times were capped at  $4 \times 10^6$  iterations; one source never arrived within that.